

be loaded by the bootloader directly. Emma can change the priority from her BIOS settings as and when she needs to change operating systems.

Bootloader initialisation

Similar to preheating the oven, the bootloader prepares the system for the OS. Once the POST sequence is successfully completed, the BIOS/UEFI firmware proceeds to load the master boot record (MBR), which contains information in the form of bytes located at the very beginning sector of a storage device such as HDD or SSD, into the RAM. The MBR contains the very first instruction called the bootloader and initialises the bootloader code to start things up. The bootloader's primary task is to locate the kernel and hand over control, setting the stage for the operating system to take over.

Loading the operating system

Finally, the cake rises, symbolising the kernel's central role in managing hardware resources and providing essential services. The bootloader activates the kernel from the storage devices into the RAM. The kernel is an interface between the applications in an operating system and the hardware interfaces such as the CPU, memory and devices, as shown in Figure 5.

The kernel handles all operating system processes such as memory management, task scheduling, I/O and overall system control. This sets up interrupt handlings and allows driver modules to be fully loaded from memory. This in turn loads up the operating system disk image from storage by mounting it and loading the drivers, much like how the cake is ready to be enjoyed after all the steps of mixing, baking, and decorating. The OS further decorates the system, allowing applications (like sprinkles) to run smoothly.

Emma's friends might never fully grasp the complexity of what goes

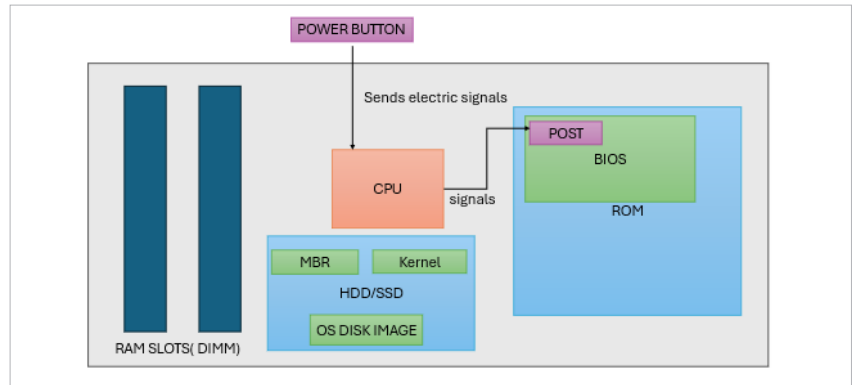


Figure 1: Workflow of the prebooting phase

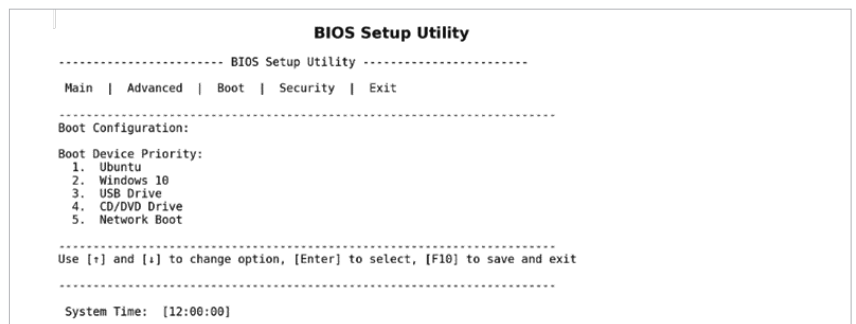


Figure 2: A sample BIOS screen

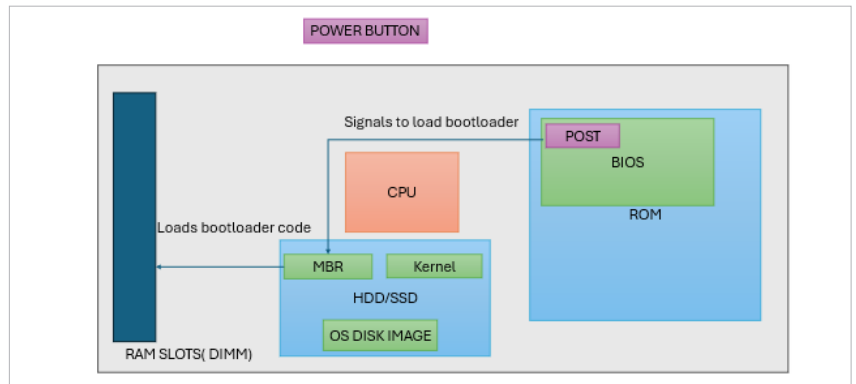


Figure 3: Workflow for loading of bootloader

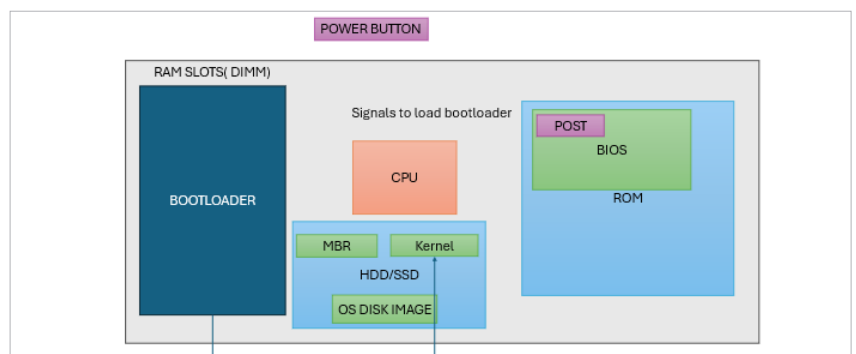


Figure 4: Workflow of bootloader activating kernel